

Comprehensive Study of a Hybrid Electric Ship for Optimum Onboard Fuel Consumption

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Abstract—With significant inroads being made in the electrification of land-based transport in recent years, there is an increased impetus on maritime transport as well for navigating towards cleaner fuels and reducing their emissions. However, the grid-distant nature of marine vessels poses several challenges for their electrification not faced by their overland counterparts. This paper addresses some of these issues and outlines the role of power electronics in tackling them through onboard/onshore converters. Moreover, a hybrid energy interface feeding the ship's propulsion system and various loads is modeled and analyzed in a simulated environment wherein an intelligent control strategy is employed to optimize the use of onboard energy resources. Finally, some future trends are highlighted in the development of electric ships for making water transport emission-free.

Index Terms—All-electric ship, electromobility, energy storage systems, power conversion, renewable energy.

I. INTRODUCTION

Marine vessels are the workhorses of the international trading industry. Globally, they are responsible for the movement of about 80% of the goods by volume [1]. Sea transport also accounts for 48% and 81% of the external trading done by the EU measured in value and volume, respectively. According to the 2021 review of maritime transport in the United Nations Conference on Trade and Development (UNCTAD), there has been a continuous growth in international maritime trade for the past decade (Fig. 1) with an exception occurring in 2020, which saw a decrease of 3.8% from 2019 due to Covid-19. Over the period 2022-2026, UNCTAD expects a growth of 2.4% in worldwide maritime trade. Based on a study by the International Maritime Organization (IMO), marine vessels are responsible for emitting 940 million tonnes of CO₂ annually and contribute to about 2.5% of global greenhouse gas (GHG) emissions [2]. With these numbers projected to go upwards, the shipping industry is facing pressure for reduction of GHG emissions. In June 2021, the IMO made several amendments to the International Convention for the Prevention of Pollution from Ships (MARPOL) to combat climate change by introducing stricter regulations on emissions that will push this sector towards becoming more energy-efficient [1].

In addition to the long-term concerns associated with the usage

This work is supported by the Marie Skłodowska-Curie Actions ETUT project (European Training network in collaboration with Ukraine for electrical Transport) and has received funding from the European Union's Horizon 2020 Research and Innovation program under grant agreement no. 955646.

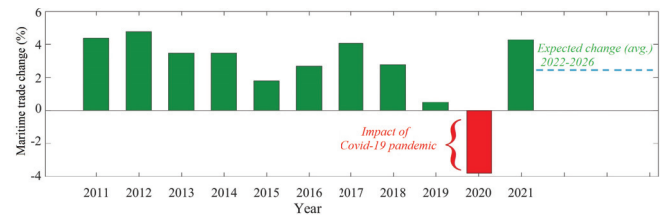


Fig. 1. International maritime trade since 2010 and its anticipated growth for the next five years (Source: UNCTAD Trade and Development Report 2021)

of conventional fuels like global warming and fuel depletion, the maritime sector also has to mind the direct consequences of fossil use. It is responsible for emitting harmful gases like sulfur oxides, nitrogen oxides, ozone, particulate matter, etc [3]. These pollutants adversely affect the health of human population as well as the flora and fauna living in the harbor areas. Therefore, a lot of research is being conducted about the applicability of bio-fuels, renewable sources, and fuel cells in the marine industry. With continuous improvements being made in battery technology [4], it is quite likely that the future of maritime transport may lie in full or hybrid electric pathways. Besides the vessel itself, in recent years, there has also been considerable interest in design of smart harbor area grids [5]. For the ships that are not battery operated, design aspects of cold ironing (a shore to ship connection when the ship is at the harbor and onboard generator is turned off to reduce emissions) are being actively evaluated. This is generally an AC/AC interface with isolation and/or frequency/voltage conversion depending upon the shore supply and the vessel's power system. In both areas, power electronics plays an essential role in optimization of available resources.

Jaster et al. [6] have modeled a green ship with AC distribution (ACD) using a fuel cell and batteries and have conducted a useful study on the cumulative gas emissions by various onboard elements under different loading conditions. However, control and anatomy of onboard converters in the model is not detailed. The integrated power system of a green ship is analyzed in [7] but the work focuses on combination of various DC power sources using dual active bridges in a DC-based distribution and does not include analysis of the propulsion system. Studies focusing on estimating emissions from fuel consumption and retrofitting and sizing a hybrid ship are also found in literature

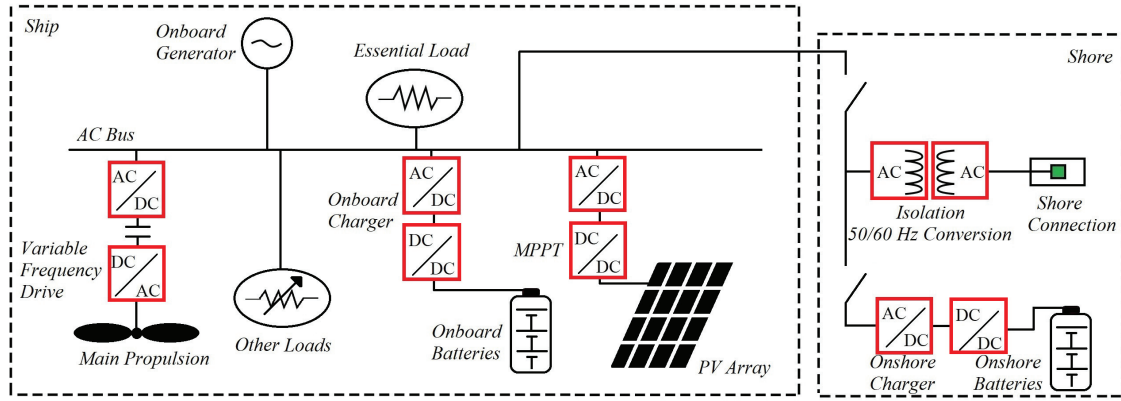


Fig. 2. Onboard hybrid energy system displaying several stages of power conversion. The onshore power supply may also have a supporting energy storage system to ensure reliable shore-to-ship power delivery in case of a weak grid for stabilization or for peak shaving

[8], [9]. In the subsequent sections of this work, a hybrid electric ship (HES) with ACD is modeled which primarily powers from an onboard diesel generator (DG). It is demonstrated that with the aid of a PV array and a battery energy storing system (BESS), the DG is operated more frequently at its optimum fuel consumption point and therefore, reduces the emissions onboard. Firstly, the models of the ship propulsion system, DG, BESS, solar array and general electrical architecture are detailed. Thereafter, the control algorithm used to optimize the onboard system is explained. Simulation results via MATLAB-Simulink validating the control method are presented which are followed by the concluding remarks.

II. ONBOARD ELECTRICAL ARCHITECTURE

Modern vessels have isolated and sophisticated power systems with elements of generation, transmission, distribution, protection, energy storage and power conversion. In addition to their transmission networks being smaller, they are distinguished from the terrestrial systems in one primary aspect: the voltage buses are not infinite and therefore, a big consumer of power onboard such as a motor drive can lead to a significant disturbance within the system. For reliability, there is often redundancy in the electrical architecture in the form of two symmetrical units for generation/transmission. A simplified single-unit diagram of a ship with redundancy omitted and a primary ACD is shown in Fig. 2. In practice, there may be additional stages of isolation, bus voltage levels, power conversion units, etc. For onboard power systems, voltages can range from 400 V in small ships to 11 kV in large vessels with total onboard power up to 100 MW. Even when docked in a port, a large vessel may have 10 MW of power requirement [10]. The distribution system on ships has been an area of extensive research in recent years. Simpler control and reduced conducting material make DC-based systems more efficient. However, the lack of a natural zero in instantaneous current makes their protection complex and expensive. With improvements being made in solid-state circuit breaker technology, DC-based systems are gaining popularity in smaller vessels in which reliable and fast protection circuits are becoming affordable for ship owners [11] while AC and hybrid systems continue to be used for larger

vessels. Based on Fig. 2, the onboard system is modeled below which consists of an 800 kW propulsion system serving as the main load. Additional AC loads can go up to 200 kW with the minimum load requirement in this study kept at 100 kW. The PV array (up to 100 kW), BESS (1000 kWh) and the DG (1000 kVA) are the power supplying elements onboard. The additional design parameters of the system are summarized in the Appendix.

A. Propulsion System

The propulsion system consists of an 800 kW induction motor driving the propeller and the converters that facilitate the motor speed control. The motor is fed from an active front end (AFE) paired with a 3- Φ inverter. In this work, closed-loop constant volts/hertz control is used for controlling motor speed and the AFE is modulated using a rotating reference frame (dq) control structure [12], aiming to minimize the reactive current drawn by the drive. A 3- Φ PLL is used to retrieve phase of the bus voltages to facilitate control of the AFE and is also employed for drawing power from the PV array and BESS. LCL filters interfacing the converters and the AC bus are deployed following a design procedure from [13]. The load torque (T) on the motor and the propeller thrust (F) are both dependent on the rotational speed of the shaft (n), density of the water (ρ), and the propeller diameter (D) [14], [15].

$$T = G_r k_q \rho D^5 n^2 \quad (1)$$

$$F = k_t \rho D^4 n^2 \quad (2)$$

wherein, G_r is the gearbox ratio, k_q is the torque coefficient and k_t is the thrust coefficient. Variations due to turbulence and cavitation are not modeled for the sake of simplicity. The net acceleration experienced by the ship is then calculated as:

$$a = \frac{(F(1 - \mu) - \lambda v^2)}{m} \quad (3)$$

wherein, m is the ship mass, μ is the thrust reduction coefficient, λv^2 is the total hull drag dependent on the ship speed, v , while the drag coefficient, λ depends upon the viscosity, water density and the wetted surface area. Substituting (2) in (3),

an expression for the ship's acceleration can be obtained. Integration of (3) yields an expression for the ship speed as:

$$v(t) = \int_0^t \frac{F(1-\mu) - \lambda v^2}{m} dt \quad (4)$$

It is pertinent to mention that (1)-(4) give a rudimentary formulation of the ship's forward motion with a single propeller and do not take into account other complex factors of propulsion, maneuvering and external thrusts that affect the ship's overall motion. However, the presented model suffices for the analysis conducted in this work.

B. BESS and PV Array

Development of onboard storage plays a key role in making the transport sector emission-free. Presently, Nickel Manganese Cobalt Oxide (Li-NMC) batteries form the most widely used chemistry in marine applications while Lithium Titanate Oxide, Nickel Metal Hydride (NMH), and solid-state electrolyte based batteries are likely to enter the maritime sector in the future [16]. Since the requirements of energy density are less stringent on ships than in terrestrial EVs, the projected future storage systems for ships are likely to differ from the latter. Additionally, reliability takes higher precedence because ships are more likely to operate in remote places where assistance might not be possible in case of a battery failure. This necessitates the presence of an emergency DG even for fully electric vessels. In this work, a 1000 kWh Li-ion battery pack, based on [17], is interfaced with a bidirectional buck-boost converter to facilitate the charge/discharge operation. Moreover, a 3- Φ inverter cascaded with an LCL filter is used to deliver/draw the power to/from the AC bus. In addition, the PV array employed comprises 47 parallel strings each consisting of 10 series-connected modules which are capable of delivering 220 W of power leading to a total capacity of 100 kW under a standard temperature of 298 K and a solar irradiance of 1 kW/m². A boost-converter, to ensure MPP tracking with the perturb and observe technique, along with an inverter is used for delivering power to the main bus by using dq control for SPWM. The BESS discharge characteristics and P-V curves of the solar panel are presented in Fig. 3.

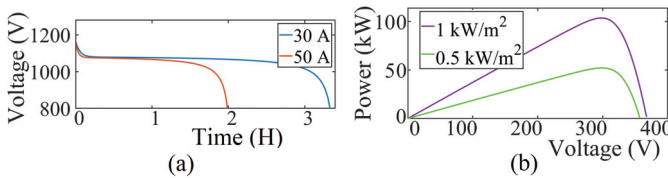


Fig. 3. (a) Discharge characteristics of onboard BESS for different values of DC current (b) P-V curve of the solar panel for different values of irradiance at 298 K

C. Additional Onboard Loads

Besides the propulsion system, a marine vessel can have several types of loads such as pumps, refrigeration units, combat systems, communication technology, radars, hotel load, etc. Depending upon the type of ship and its state of operation like navigation, floating or in combat, any of these can be in use

and demand the required power from the onboard system. To emulate this, the model employs a constant load of 100 kW (labeled as essential load in Fig. 2) and a variable load (labeled as other loads in Fig. 2) capable of drawing both real and reactive power is used. The active and reactive power demanded by the variable load are capped at 100 kW and 50 kVARs, respectively.

D. Onboard Generator

Unless the ship is operating at a significantly low speed, the onboard DG is responsible for delivering bulk of the power to various loads. A simplified synchronous machine, with its rotation governed by the following expression, is used to represent the DG.

$$\Delta\omega(t) = \int_0^t \frac{T_m - T_e - k_d \Delta\omega(t)}{2H} dt \quad (5)$$

wherein, $\Delta\omega$ is the variation in speed, H is the inertia constant, k_d is the damping factor, T_m and T_e are the mechanical and electromagnetic torques, respectively. It is desired that the DG is operated at 70% of the maximum capacity where it is specified to have the least specific fuel consumption (SFC) which is a measure of the amount of fuel consumed per unit of energy that is generated. There are models in literature that estimate emissions from the diesel generator based on the generator power, dynamic fuel consumption characteristics and operating data [6]. A typical SFC vs power output curve is shown in Fig. 4 indicating that a deviation from the optimum point is undesirable and leads to inefficient operation.

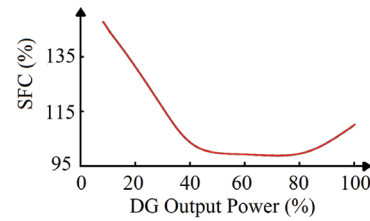


Fig. 4. Variation of SFC with the power generated for a typical DG

III. CONTROL STRATEGY

The onboard system consists of several units that can significantly vary their power output or consumption. In addition to the localized control structures that are responsible for converter switching, it is paramount to incorporate a central energy management system which allocates the power to be supplied by the sources on board to meet the load demand. Depending on the SoC of battery, available solar power (P_{PV}), and the total load demand (P_{load}), the energy requirement from onboard resources can be allocated as illustrated in Fig. 5. The power allocation algorithm has one primary objective, which is to maximize the use of DG (P_{DG}) around its optimum point of operation ($P_{DG,opt}$) and to avoid low-efficiency zones while simultaneously using the onboard solar power to its full capacity. This optimization is divided into three modes based

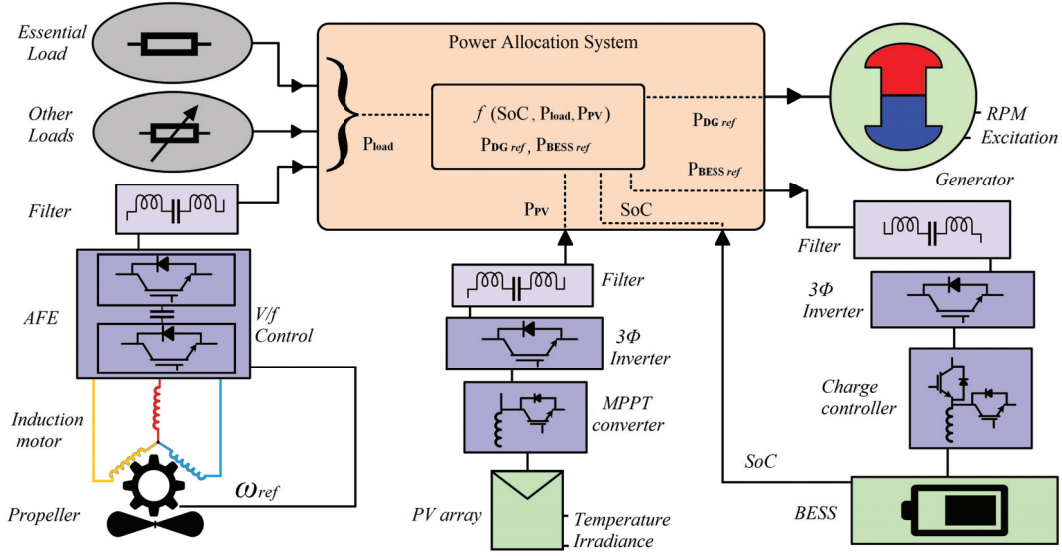


Fig. 5. Block diagram representing the complete control structure of the onboard electrical system. The specifications related to the various power converters, load and source parameters are summarized in the appendix.

on the battery SoC. Consequently, the power demanded from the battery (P_{BESS}) may be positive (discharging) or negative (charging) depending on the load requirement and available PV power.

Case I: $SoC \leq SoC_{min}$

In this mode, depending upon the load requirement and available solar power, the DG and PV array deliver power to the load and may charge the battery according to the following scenarios:

- If $P_{load} \leq P_{PV}$, then the excess solar energy is used to charge the BESS and the DG is turned off.
- If $P_{PV} < P_{load} \leq P_{PV} + P_{DG,opt}$, then the DG and PV array feed the load and charge the battery.
- If $P_{PV} + P_{DG,opt} < P_{load} < P_{PV} + P_{DG,max}$, then the DG and PV array only feed the load and excess deviation of P_{DG} from $P_{DG,opt}$ is avoided.

Case II: $SoC_{min} < SoC < SoC_{max}$

In this mode, the BESS may act as a load or source to drive the DG nearer towards its optimum point of operation when it is running or allow it to be turned off if possible.

- If $P_{load} \leq P_{PV}$, then the excess solar energy is used to charge the BESS while the DG is turned off.
- If $P_{PV} < P_{load} \leq P_{PV} + P_{BESS,nominal}$, then the BESS and PV array feed the load and the DG is still off.
- If $P_{PV} + P_{BESS,nominal} < P_{load} \leq P_{PV} + P_{DG,opt}$, then the DG and PV array feed the load and charge the BESS.
- If $P_{PV} + P_{DG,opt} < P_{load} < P_{PV} + P_{DG,max}$, then the BESS, DG and PV array feed the load.

Case III: $SoC \geq SoC_{max}$

When this mode is attained, the battery can no longer be charged in order to avoid damage and is only allowed to act as a source of power.

- If $P_{load} \leq P_{PV}$, then only the required solar power is delivered to the load.
- If $P_{PV} < P_{load} \leq P_{PV} + P_{BESS,nominal}$, then the BESS and PV array feed the load and the DG remains off.
- If $P_{PV} + P_{BESS,nominal} < P_{load} \leq P_{PV} + P_{DG,opt}$, then only the DG and PV array feed the load.
- If $P_{PV} + P_{DG,opt} < P_{load} < P_{PV} + P_{DG,max}$, then the BESS, DG and PV array feed the load.

IV. RESULTS AND DISCUSSION

The system performance is examined under various operating conditions via MATLAB-Simulink simulated with a sampling interval of 1 μs . Fig. 6 shows the initial response of the system near the maximum loading condition when the ship is starting from rest. In this scenario, the DG is operated at 80% of its maximum capacity while the PV array and BESS supply 100 kW and 50 kW of power, respectively. Corresponding to the steady-state in Fig. 6, the AC bus voltage, inverter currents injected from the PV Array and BESS are shown in Fig. 7. The power is delivered to the bus at a near unity input power factor validating dq-frame-based PWM control.

From (1) it can be inferred that the propulsion power varies with the cube of rotational speed. Therefore, for lower speed operation, the total onboard load reduces significantly. Such a condition is illustrated in steady-state via Fig. 8. Since the DG power requirement is less than optimum, the AC bus delivers power to the battery to operate the DG at a higher efficiency.

When the ship is stationary, the load demand on the ship is significantly reduced as the propulsion system which is the primary consumer of power becomes inactive. Fig. 9 shows the system operation when the total onboard demand comprises the essential load of 100 kW and an additional 50 kW of other loads. Under these conditions, the DG is no longer required to supply power to the bus and the load is shared between

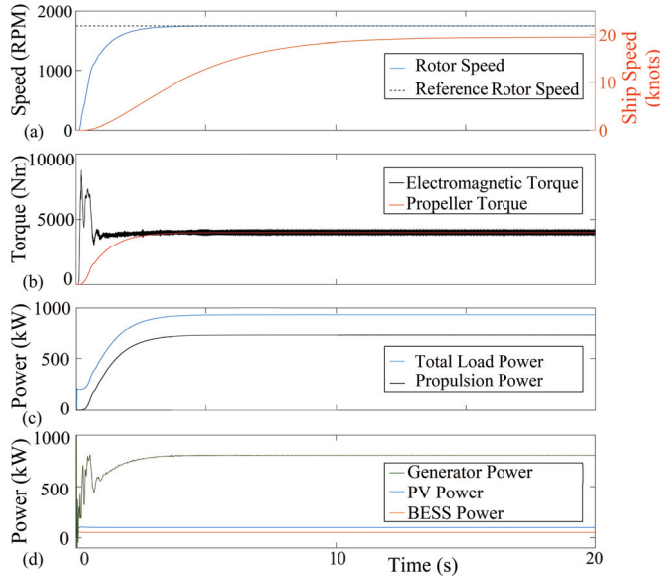


Fig. 6. System response at maximum loading conditions: (a) Motor response and ship speed for a speed reference of 1750 RPM (b) Electromagnetic torque and the propeller load torque (c) Load power distribution (d) Power supplied by DG, PV array (at irradiance 1 kW/m² and temperature 298 K), and BESS (at 50% SoC and nominal discharge current).

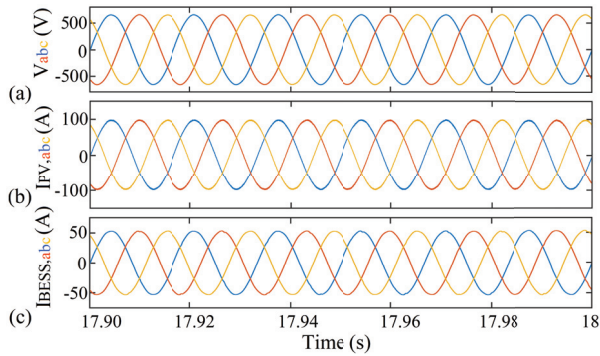


Fig. 7. Steady-state system performance: (a) AC bus 3-Φ voltages (phase) (b) Injected 3-Φ currents from the PV system (c) Injected 3-Φ currents from the BESS.

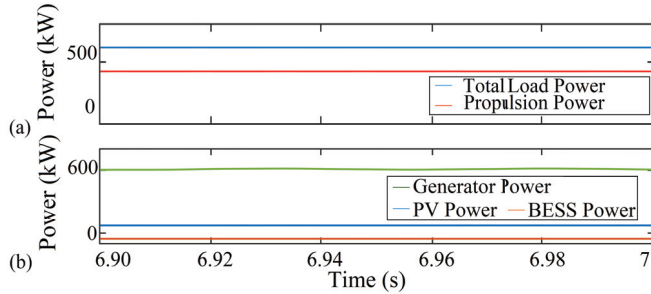


Fig. 8. System response for medium loading conditions: (a) Load power distribution for a speed reference of 1450 RPM (b) Power supplied by DG, PV array (at irradiance 0.75 kW/m² and temperature 298 K), and BESS (negative power indicates charging of the BESS).

the PV array and BESS. Moreover, since the AC bus is no longer fed from DG, the BESS inverter takes up the role of

forming the grid while previously following it in the presence of DG. In the case of larger AC distribution systems that employ several distributed storage elements, synchronization, power-sharing and seamless transition from grid-forming to grid-following modes by these units becomes essential. In order to facilitate this, droop control is widely employed in these converters with a secondary control mechanisms to avoid post-transition oscillations [18].

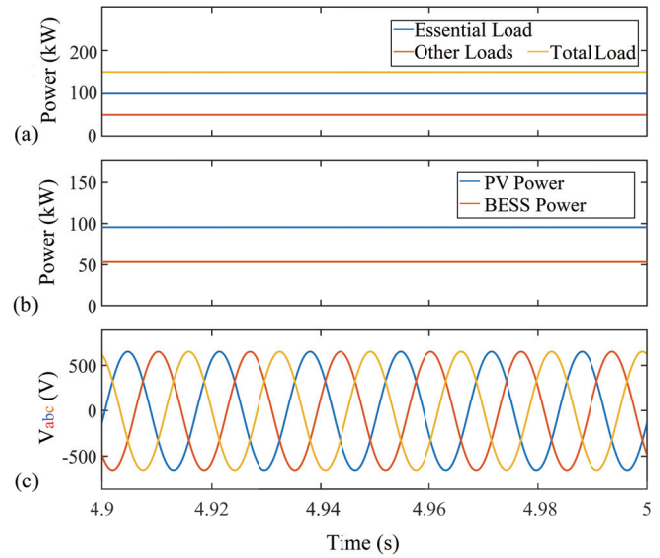


Fig. 9. System response during stationary operation: (a) Load power distribution (b) Power supplied by PV array (at irradiance 1 kW/m² and temperature 298 K), and BESS. (c) AC bus voltage set up by the 3-Φ inverter

The load curve analyzed in this work based on the preceding three scenarios is shown in Fig. 10. In accordance with Fig. 4, the load curve is used to estimate the total fuel consumption for the three cases. This data is presented in Table I. It is observed that the use of a hybrid energy interface enables the system to avoid high fuel consumption zone for the DG. The cumulative fuel consumed in this case is about 2.07 units which is 27% less than in the conventional DG-only-ship (2.845 units). A significant portion of this difference arises from the turn-off duration of DG in case of the hybrid ship.

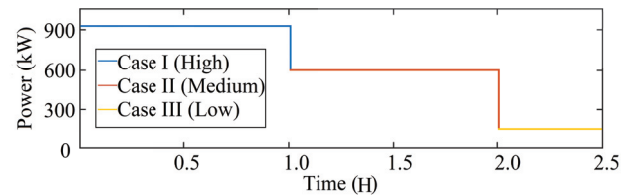


Fig. 10. Load curve showing the durations of high power consumption (blue), medium power consumption (red) and low power consumption (yellow)

TABLE I
LOAD POWER, DISTRIBUTION OF DG POWER AND SFC CORRESPONDING
TO THE LOAD CURVE IN FIG. 10. THE ASTERISK (*) INDICATES THE
CONVENTIONAL SCENARIO WHERE ONLY DG IS USED ONBOARD

	Case I	Case II	Case III
P_{load} (kW)	950	625	150
P_{DG} (pu)	0.800	0.600	0
P_{DG*} (pu)	0.950	0.625	0.150
SFC (pu)	1.02	1.05	0
SFC* (pu)	1.10	1.02	1.45
Duration (H)	1.0	1.0	0.5
Total consumption (units)	1.02	1.05	0
Total consumption * (units)	1.10	1.02	0.725

V. CONCLUSION

In this work, a case study of a hybrid electric ship was carried out wherein it was demonstrated that the use of onboard energy storage and PV power can help in optimal usage of the diesel generator and thereby reduce its specific fuel consumption and emission of greenhouse gases. Compared to a conventional DG based ship, a reduction of 27% in fuel consumption is achieved using the hybrid architecture. As energy demands across the world continue to soar amidst the decline in available fuel reserves, it is imperative for the transport sector to adapt to alternative energy resources. For the maritime sector in particular, which is at the heart of international trade, there is a lot of ground still to be covered in making future ships sustainable and completely emission-free. The use of alternative fuels, storage systems and power distribution methods is being continuously explored. There is a growing trend of ship power systems becoming similar to terrestrial micro-grids and incorporating elements of onboard generation, storage, and alternative fuels for more efficient energy utilization, and power electronics plays a key role in their amalgamation. Unlike for terrestrial EVs, where plugging in is possible virtually anywhere due to the domestic power supply availability, fleet charging needs to be organized strategically. Therefore, establishment of standards, schedules and government policies will go a long way in helping address the problems at the ports.

APPENDIX

- Propulsion System: Nominal Power = 800 kW; Voltage (line to line) = 800 V (RMS); Frequency = 60 Hz; Poles = 4; Propeller Gear Ratio = 1/5; Torque Coefficient = 0.06; Water Density = 1026 kg/m³; Propeller Diameter = 0.825 m; Thrust Coefficient = 0.5; Hull Drag Coefficient = 300 kg/m; Ship Mass = 100,000 kg; Thrust Reduction Factor = 0.2
- BESS: Nominal Voltage = 1000 V; Rated Capacity = 100 Ah; Nominal Discharge Current = 50 A; Cut-off Voltage = 750 V; Fully Charged Voltage = 1150 V
- PV Module: Maximum Power = 220.22 W; Open Circuit Voltage = 36.8 V; Short Circuit Current = 8 A; Temperature Coefficient of Voltage = - 0.3651%/deg.C; Temperature Coefficient of Current = 0.075%/deg.K; Shunt Resistance = 145 Ω ; Series Resistance = 0.33 Ω

- Power Converters: Switching Frequency = 10 kHz
- Generator: Nominal Motor Power = 1000 kVA; Voltage (line to line) = 800 V (RMS); Frequency = 60 Hz; Poles = 4; Total Moment of Inertia = 2 kgm²

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